CSE 400 Lecture 6 Scribe

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Generate a lecture scribe that will serve as exam-oriented reference material for my course : CSE 400.
Use only the provided context (lecture slides and relevant textbook material for Lecture 6).
Include:
all definitions and notation introduced in the lecture
stated assumptions and conditions
theorem or result statements
proofs or derivations only if they appear in the lecture, written step by step
worked examples only if they were covered in the lecture, with intermediate steps
Do not introduce new explanations, intuition, examples, or results.
Do not go beyond the scope of the provided context.

Organize the scribe clearly with appropriate sections so it can be reliably used by me for exam revision.

CSE400 — Lecture 6 (Discrete RVs, Expectation and Problem Solving) — Scribe (exam-oriented)

Source: Lecture slides (Lecture 6).

Lecture / Instructor (as shown in slides): Dhaval Patel, PhD — SEAS-Ahmedabad University

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1. Scope (as given on the slides)

Items covered in this lecture (explicitly from the slides):

- Types of discrete random variables (Bernoulli, Binomial, Geometric, Poisson) — listed.
- **Probability Mass Function (PMF)** — definition and properties.

- Examples: tossing 3 fair coins (PMF example); a discrete PMF given by $p(i) = c\frac{\lambda^i}{i!},\ i = 0, 1, 2, \ldots$ (Poisson form) — find normalizing constant and certain probabilities (worked on slides).
- **Bayes' Theorem** — statement (recap) and two worked examples (auditorium rows; communication system — latter introduced).
- (Expectation, CDF/PDF, and moments are listed in the outline but proofs/derivations for expectation do not appear on the visible slides; only PMF and Bayes examples contain stepwise work on the provided pages.)

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2. Definitions & Notation (as introduced on slides)

- **Random Variable (RV)** — (slide heading / motivation): a variable that assigns real values to outcomes of an experiment (visuals shown).
  No additional formal definition text beyond slides' illustrations.
- **Discrete random variable** (slide):
  - Range (support) $R_X = \{x_1, x_2, x_3, \ldots\}$ (finite or countably infinite).
  - Each possible value has strictly positive probability (probabilities assigned to single values).
  - Notation: $P_X(x_k) = P(X = x_k)$ for $k = 1, 2, 3, \ldots$.
- **Probability Mass Function (PMF)** — (slide text):
  For a discrete RV X with range $R_X = \{x_k\}$, the function

$$p_X(x_k) = P(X = x_k), \qquad k = 1, 2, 3, \ldots$$

  is called the PMF of X.
- **PMF property (normalization)** — (slide):

$$\sum_k p_X(x_k) = 1.$$

- **Bayes Formula (recap on slide)** — boxed formula from slides:

$$P(B_i \mid A) = \frac{P(A \mid B_i)\,P(B_i)}{\sum_j P(A \mid B_j)\,P(B_j)}.$$

  (Also the simple conditional relation used: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.)

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3. Worked example — Tossing 3 fair coins (slide images/pages 8–10)

Set-up (from slide): Toss 3 fair coins. Let Y denote the number of heads. Range: $\{0, 1, 2, 3\}$.

PMF values (as on slide):

- $P(Y = 0) = P(t, t, t) = \frac{1}{8}$.
- $P(Y = 1) = P(\{t, t, h\}, \{t, h, t\}, \{h, t, t\}) = \frac{3}{8}$.
- $P(Y = 2) = P(\{h, h, t\}, \{h, t, h\}, \{t, h, h\}) = \frac{3}{8}$.
- $P(Y = 3) = P(h, h, h) = \frac{1}{8}$.

Normalization check (as on slide):

$$1 = P\left(\bigcup_{i=0}^{3} \{Y = i\}\right) = \sum_{i=0}^{3} P(Y = i).$$

(These exact probabilities and the summation equality appear on the lecture slides.)

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4. Worked example — PMF of the form $p(i) = c\frac{\lambda^i}{i!}$, $i = 0, 1, 2, \ldots$ (Poisson form) — slides 14–16

Given (slide): The PMF of a random variable X is $p(i) = c\frac{\lambda^i}{i!}$ for $i = 0, 1, 2, \ldots$, where $\lambda > 0$. Find $P(X = 0)$ and $P(X > 2)$.

Step-by-step (as on slides):

1. Use normalization $\sum_{i=0}^{\infty} p(i) = 1$:

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1.$$

2. Recognize the power series for the exponential (slide shows this identity):

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}.$$

3. Therefore:

$$c\,e^{\lambda} = 1 \qquad \Rightarrow \qquad c = e^{-\lambda}.$$

4. Compute $P(X = 0)$ (slide):

$$P(X = 0) = c\frac{\lambda^0}{0!} = e^{-\lambda}.$$

5. Compute $P(X > 2)$ (slide shows the complementary summation):

$$P(X > 2) = 1 - P(X \leqslant 2) = 1 - (P(X = 0) + P(X = 1) + P(X = 2)).$$

   Substitute the PMF terms with $c = e^{-\lambda}$:

$$\begin{aligned} P(X > 2) &= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda}) \\ &= 1 - e^{-\lambda}(1 + \lambda + \tfrac{\lambda^2}{2}). \end{aligned}$$

(The slides present these steps in the same order; the derivation above follows the slide content exactly.)

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5. Bayes' Theorem — statement and worked auditorium example (slides ~17–21)

5.1 Bayes' Theorem (recap — slide)

- Statement (as on slide):

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}.$$

- General (as boxed on slide):

$$P(B_i \mid A) = \frac{P(A \mid B_i)\,P(B_i)}{\sum_j P(A \mid B_j)\,P(B_j)}.$$

5.2 Example #1 — Auditorium with 30 rows (slides)

Problem statement (from slide):

- Auditorium has 30 rows. Row 1 has 11 seats, Row 2 has 12 seats, Row 3 has 13 seats, … up to Row 30 which has 40 seats (the slide lists row seat counts progressively to Row 30 = 40).
- A door prize is given by: first randomly select a row (each of the 30 rows equally likely), then randomly select a seat within that row (each seat in the chosen row equally likely).

Questions shown on slide:

1. Compute the probability that seat 15 (denote event S_{15}) is selected.
2. Compute $P(S_{15} \mid R_{20})$ (the probability seat 15 is selected given row 20 selected).
3. Compute $P(R_{20} \mid S_{15})$ (the posterior probability that row 20 was selected given that seat 15 was selected).

Work shown on slides (steps and formulas):

- Step (1) — total probability for S_{15} (slide shows):

$$P(S_{15}) = \sum_{k=1}^{30} P(S_{15} \mid R_k) P(R_k).$$

  (Here $P(R_k) = 1/30$ for each row since the row selection is equally likely; $P(S_{15} \mid R_k)$ equals $1/$ (number of seats in row k) if seat 15 exists in that row, otherwise 0.)
- The slide shows the numeric evaluation of the above sum and records the result:

$$P(S_{15}) = 0.0342 \quad \text{(value shown on slide).}$$

- Step (2) — conditional probability (slide):
  For a specific row R_{20}, if seat 15 is present in row 20, then

$$P(S_{15} \mid R_{20}) = \frac{1}{\text{(number of seats in Row 20)}}.$$

  (This formula is explicitly used on the slide.)
- Step (3) — Bayes inversion (slide gives formula and computes numerically):

$$P(R_{20} \mid S_{15}) = \frac{P(S_{15} \mid R_{20}) P(R_{20})}{P(S_{15})}.$$

  The slide shows the substitution and a numeric result (as appearing on the slide):

$$P(R_{20} \mid S_{15}) = \frac{\left(\frac{1}{\text{seats in Row 20}}\right) \cdot \frac{1}{30}}{0.0342} \quad \text{(numeric evaluation shown on slide}$$

> **Note:** The slide presents the explicit total-probability summation for $P(S_{15})$, the formula $P(R_{20} \mid S_{15}) = \frac{P(S_{15} \mid R_{20}) P(R_{20})}{P(S_{15})}$, and the numeric values as shown above (the slide shows $P(S_{15}) = 0.0342$ and performs the Bayes substitution).

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6. Other items explicitly present on slides (no derivation given on these slides)

- **Discrete vs Continuous variable** — slide bullets (discrete: countable support, PMF; continuous: uncountable support, PDF). (These items are presented on slides; no further derivation was given on the provided pages.)
- **CDF / PDF** — listed in the outline (Definition, properties and examples are in the lecture outline); explicit PMF/PDF distinction figure appears on slides.
- **Expectation, linearity, moments** — listed in the lecture outline (Expectation of RVs; expectation of a function of RV; linearity; nth moments / central moments: variance, skewness, kurtosis). *No step-by-step derivations for expectation or moments are present on the visible pages of the provided Lecture 6 slides.* Therefore no extra derivations are included here beyond what the slides show.

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7. Items explicitly shown as boxed / theorem statements on slides

- **Bayes Formula [Proposition 3.1]** — boxed on slide:

$$P(B_i \mid A) = \frac{P(A \mid B_i)\,P(B_i)}{\sum_j P(A \mid B_j)\,P(B_j)}.$$

(That boxed formula is included verbatim from the slides.)

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8. What the scribe does not add

Per your instructions and per the slides: **no additional explanations, intuition, new examples, or results** beyond the exact material, definitions, formulas, derivations, and worked steps that appear in Lecture 6 slides were added.

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9. Quick reference checklist (for exam revision) — *exact expressions and facts appearing on slides*

- PMF definition: $p_X(x_k) = P(X = x_k)$.
- PMF normalization: $\sum_k p_X(x_k) = 1$.

- Example (3 fair coins): $P(Y = 0) = \frac{1}{8}$, $P(Y = 1) = \frac{3}{8}$, $P(Y = 2) = \frac{3}{8}$, $P(Y = 3) = \frac{1}{8}$.
- Poisson-form PMF normalization: if $p(i) = c\frac{\lambda^i}{i!}$ then $c = e^{-\lambda}$. Hence $P(X = 0) = e^{-\lambda}$ and $P(X > 2) = 1 - e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2})$.
- Bayes formula (boxed).
- Auditorium example: $P(S_{15}) = \sum_{k=1}^{30} P(S_{15} \mid R_k)P(R_k)$ (slide gives numeric = 0.0342 on slide); and $P(R_{20} \mid S_{15}) = \dfrac{P(S_{15} \mid R_{20})P(R_{20})}{P(S_{15})}$ (slide shows substitution and numeric evaluation).

If you want, I can extract and place the exact slide images or produce a single-page printable scribe that *only* contains the equations and boxed formulas from Lecture 6 (no added text). (I will not add any new explanation beyond the slides.)